

TASK 2: PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Cyber Security Internship – Task 2

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**Task:
Phishing Email Detection & Awareness**

**Project Focus:
Identification, Analysis and Classification of Phishing Emails**

**Examples Analysed:
4 Email Samples**

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1. INTRODUCTION

Phishing is a common form of cyber attack in which attackers attempt to deceive users through fraudulent emails, messages, websites, or other communication methods. These attacks often imitate trusted organizations or services and use social-engineering techniques to encourage users to click links, provide information, or make payments.

This task focuses on the identification and analysis of phishing-related email examples. Four different email samples were examined to identify suspicious characteristics, assess their risk, and determine appropriate user actions.

The analysis considers factors such as sender information, email content, branding, unexpected requests, links, social-engineering techniques, requests for personal information, and payment-related activity.

The purpose of this exercise is not only to identify suspicious emails but also to understand how phishing techniques can influence users and how security awareness can help prevent successful attacks.

2. OBJECTIVES

The main objectives of this task are:

1. To identify phishing characteristics present in different email samples.
2. To analyse suspicious email content and identify indicators such as unexpected requests, suspicious links, impersonation, payment requests, and social-engineering techniques.
3. To classify email samples according to their apparent risk level, such as safe, suspicious, or phishing.
4. To assess the potential risks associated with interacting with suspicious emails.
5. To recommend appropriate security actions that users should take when they receive suspicious or unexpected emails.
6. To improve phishing awareness by understanding common techniques used to manipulate users into clicking links, sharing information, or making payments.

7. To develop practical security awareness skills that can help users identify and avoid phishing attacks.

3. METHODOLOGY

The phishing email analysis was carried out using a structured and evidence-based approach. Four email samples were reviewed individually, and the visible characteristics of each message were examined to identify possible phishing indicators.

The following steps were followed during the analysis:

3.1 Collection of Email Evidence

Screenshots of the selected email examples were collected and stored in the project's **Screenshots** folder. The screenshots were used as visual evidence during the analysis.

3.2 Initial Email Review

Each email was reviewed to understand its purpose, message theme, sender information, subject, branding, and the action requested from the recipient.

3.3 Identification of Phishing Indicators

The emails were examined for common warning signs, including:

- Unexpected messages or notifications
- Brand impersonation
- Suspicious or external links
- Requests for personal information
- Payment requests
- Urgent or pressure-based language
- Social-engineering techniques
- Unexpected attachments or document notifications

3.4 Risk Assessment

After identifying the relevant indicators, each email was assigned an appropriate risk level based on the available evidence.

The four examples were classified as:

- **Example 1:** Phishing — High Risk
- **Example 2:** Safe / Legitimate-looking based on available evidence — Low Risk
- **Example 3:** Suspicious / Potential Phishing — Medium Risk
- **Example 4:** Phishing — High Risk

3.5 Recommended Security Actions

For each example, appropriate user actions were identified. These included avoiding suspicious links, verifying unexpected communications through trusted channels, protecting personal information, avoiding unverified payments, and reporting suspicious emails.

3.6 Documentation

A separate analysis document was prepared for each email example. The detailed analyses and screenshots were organized in the project's Analysis and Screenshots folders respectively.

4. ANALYSIS OF EMAIL EXAMPLES

4.1 Example 1 – SharePoint/Fax Notification

Classification: Phishing

Risk Level: High

The first example is a fake fax/document notification that uses the appearance of a trusted business service. The email encourages the recipient to click a “Print & Preview Here” link to access the supposed fax.

Important indicators include brand impersonation, an unexpected document notification, a suspicious link, generic recipient information, and social-engineering techniques designed to create curiosity and encourage interaction.

The email was therefore classified as a Phishing Email with High Risk.

4.2 Example 2 – PayPal Account Notifications

Classification: Safe / Legitimate-looking based on available evidence

Risk Level: Low

The second example contains two PayPal-related notifications: a “Welcome to PayPal” message and a verification-code notification.

The visible sender domains, content, and branding do not show obvious phishing indicators. However, because the analysis was based on screenshots rather than complete email headers and authentication results, the emails were classified as legitimate-looking based on available evidence, rather than being declared completely genuine.

4.3 Example 3 – Punchbowl Invitation

Classification: Suspicious / Potential Phishing

Risk Level: Medium

The third example is an online invitation that encourages the recipient to click an “Open Invitation” button.

The main concern is that the message creates a clear call to action and may use curiosity or an unexpected invitation as a social-engineering technique. The screenshot does not provide sufficient technical evidence to conclusively confirm that the email is malicious.

It was therefore classified as Suspicious / Potential Phishing with Medium Risk.

4.4 Example 4 – Free Items / Logistics Fee

Classification: Phishing

Risk Level: High

The fourth example offers several supposedly free items to selected students but requires the recipient to pay a \$150 logistics fee for each item.

The email also requests personal information and provides payment options. The combination of an unexpected valuable offer, an upfront fee, personal-information requests, and pressure created by limited availability represents strong phishing and advance-fee scam indicators.

The email was therefore classified as a Phishing Email with High Risk.

Example	Email Type	Classification	Risk Level
Example 1	Fake Fax / Document Notification	Phishing	High
Example 2	PayPal Account Notifications	Safe / Legitimate-looking	Low
Example 3	Online Invitation	Suspicious / Potential Phishing	Medium
Example 4	Free Items / Logistics Fee	Phishing	High

This table is useful because it lets someone reviewing your internship task understand the overall result immediately.

5. EMAIL EVIDENCE

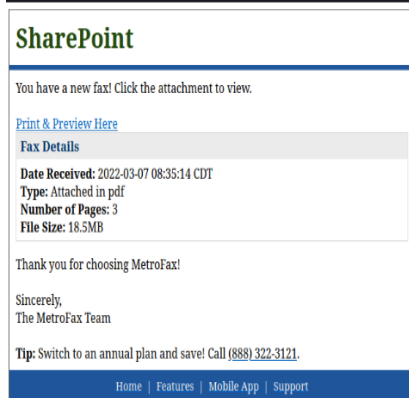
5.1 Example 1 – SharePoint/Fax Notification

Classification: Phishing

Risk Level: High

Evidence

Screenshot:



.....

This message may contain confidential information. If you are not the intended recipient please:
1) inform the sender that you have received the message in error before deleting it; and
1) do not disclose, copy or distribute information in this e-mail or take any action in relation to its content (to do so is strictly prohibited and may be unlawful).
Thank you for your co-operation.

NHSmil is the secure email, collaboration and directory service available for all NHS staff in England. NHSmil is approved for exchanging patient data and other sensitive information with NHSmil and other accredited email services.

For more information and to find out how you can switch visit [Joining NHSmil - NHSmil Support](#)

Brief observation:

The email uses a trusted-service appearance and asks the recipient to interact with a “Print & Preview Here” link.

5.2 Example 2 – PayPal Notifications

Classification: Safe / Legitimate-looking

Risk Level: Low

Evidence

Screenshot:



Brief observation:

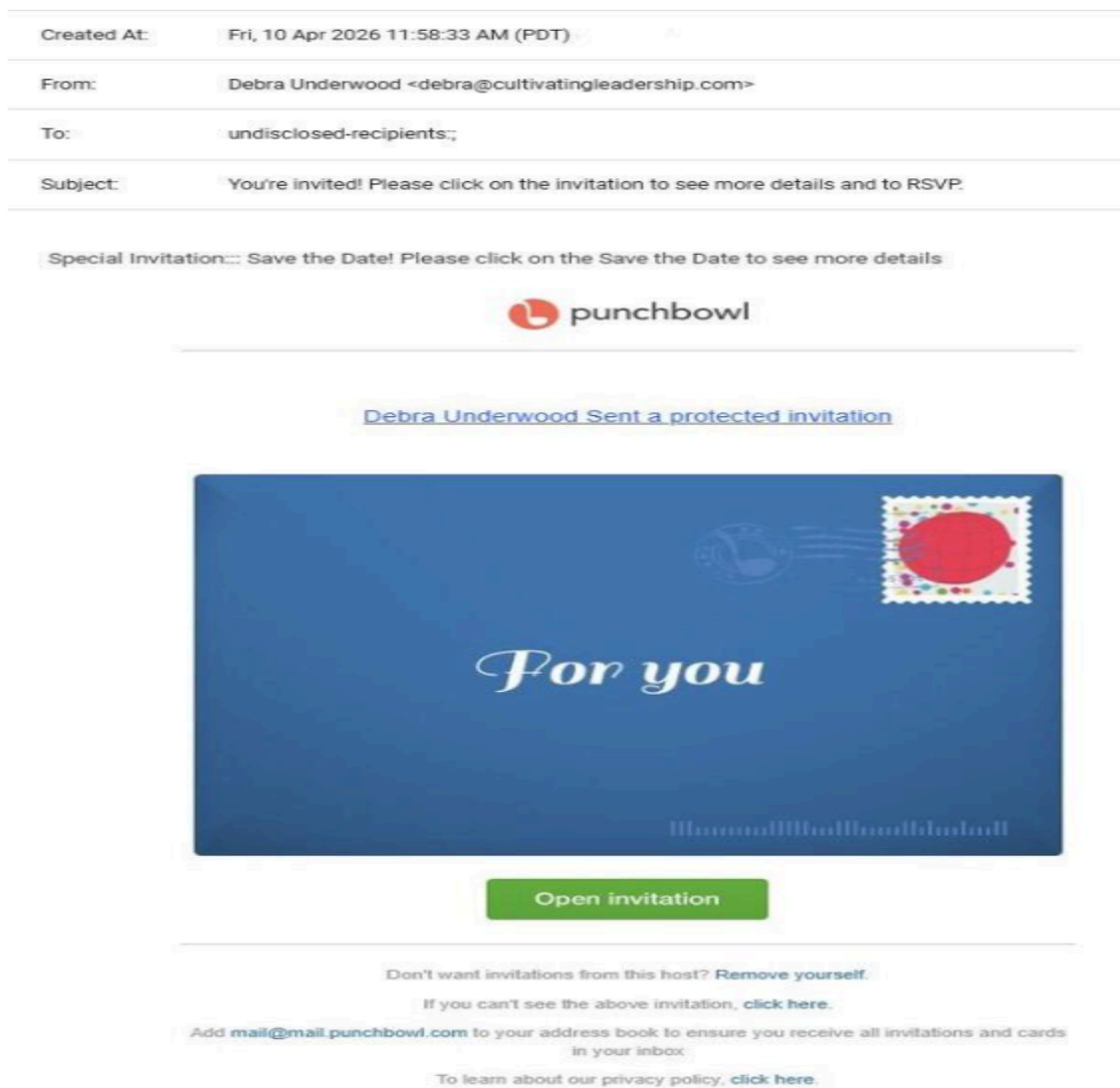
The screenshot contains PayPal account-related notifications. No obvious phishing indicators are visible from the screenshot alone.

5.3 Example 3 – Punchbowl Invitation

Classification: Suspicious / Potential Phishing

Risk Level: Medium

Evidence Screenshot:



Brief observation:

The email uses an invitation and an “Open Invitation” button as the main call to action.

5.4 Example 4 – Free Items / Logistics Fee

Classification: Phishing

Risk Level: High

Evidence Screenshot:

Dear Selected Students,

We're excited to share a special opportunity made possible by **Mrs. Maria**, a cherished retired music educator who spent over three decades nurturing the creativity and talent of young musicians. As she prepares to relocate to be closer to family, Mrs. Maria has decided to part with several personal items once treasured by both herself and her late husband, a fellow musician and passionate advocate for the arts.

Her wish is that these items continue to inspire and support students who will truly appreciate and use them.

 **Items Available for Gifting:**

- **Violin** – Yamaha AV7-44SG
- **Guitar** – Martin 000-42 (Eric Clapton Signature Edition)
- **Keyboard** – Roland D-50 Synthesizer
- **Digital Camera** – Leica S (Typ 006)
- **Drone** – Mavic Quadcopter

These items are being offered free of charge. However, a **logistics fee of \$150 per item** is required to cover:

- Professional-grade packaging
- Full-value shipping insurance
- Tracked and secure delivery

Please note: Due to mobility limitations, in-person pickup is not possible.

 **Requesting More Than One Item?**

If you have genuine interest and can cover the logistics cost, you're welcome to request more than one item, subject to availability.

 **About Mrs. Maria:**

Mrs. Maria taught music at both elementary and high school levels, leaving a lasting impact on countless students through her passion, patience, and dedication. This gift reflects her lifelong commitment to the arts and honors the legacy of her late husband.

 **How to Request:**

To express your interest, **please text Mrs. Maria directly at the number provided in the subject line. Do not reply to this email.**

In your message, include:

- Your **full name**
- Your **complete shipping address**
- Your preferred **payment method** for the logistics fee (Apple Pay, Zelle, or as instructed)
- The **item(s)** you're requesting

 **Important Note:**

This message has been sent to a small, carefully selected group of students. It will not be re-sent. If interested, please **save or screenshot** this message for your reference.

We are deeply grateful to Mrs. Maria for her incredible generosity and hope these meaningful items find new homes where they will

Brief observation:

The email offers supposedly free items but requests a logistics fee and personal information, which are strong warning signs.

6. COMMON PHISHING INDICATORS

The analysis of the four email examples identified several common characteristics that can help users recognize potentially malicious or suspicious messages.

6.1 Brand Impersonation

Attackers may use the name, branding, or visual appearance of trusted organizations to make an email appear legitimate. Example 1 demonstrates this technique by presenting a fax notification using a trusted business-service appearance.

6.2 Suspicious or Unexpected Links

Links and buttons can be used to direct users to potentially unsafe destinations. Example 1 contains a “Print & Preview Here” link, while Example 3 uses an “Open Invitation” button as its main call to action.

6.3 Unexpected Messages

An unexpected notification, document, invitation, or account message can be used to create curiosity and encourage a user to interact with an email.

Examples 1 and 3 demonstrate how unexpected fax notifications and invitations can be used as potential social-engineering lures.

6.4 Requests for Personal Information

Phishing and scam emails may attempt to collect personal information from recipients. Example 4 requests information such as the recipient's name, address, preferred payment method, and selected item.

6.5 Payment Requests

Unexpected payment requests are an important warning sign. Example 4 requests a \$150 logistics fee for each item, despite initially presenting the items as free.

6.6 Social Engineering

Social engineering attempts to influence users psychologically rather than relying only on technical methods.

The examples demonstrate different approaches, including:

- Curiosity about an unexpected fax
- Curiosity about an invitation
- Attraction to supposedly free valuable items
- Pressure created by limited availability

These techniques can encourage recipients to act before properly verifying the message.

6.7 Professional-Looking Appearance

A phishing email may use professional formatting, familiar branding, or business-style language to appear trustworthy. Example 1 demonstrates that a professional-looking design should not automatically be considered proof of authenticity.

Phishing Indicator	Example(s)	Risk/Concern
Brand impersonation	Example 1	Can create false trust
Suspicious link/button	Examples 1 & 3	May lead to unsafe destination
Unexpected notification	Examples 1 & 3	Can trigger curiosity

Personal information request	Example 4	Privacy/security risk
Payment request	Example 4	Possible financial scam
Social engineering	Examples 1, 3 & 4	Manipulates user behaviour
Professional appearance	Example 1	Can make fraudulent email appear genuine

7. RISK ASSESSMENT AND COMPARISON

7.1 Risk Assessment Approach

Each email was assessed according to the number and seriousness of suspicious indicators identified during the analysis. Factors considered included suspicious links, impersonation, unexpected requests, requests for personal information, payment demands, and social-engineering techniques.

The assessment was based on the evidence available in the email screenshots and should not be considered a substitute for complete technical email-header or link analysis.

7.2 Risk Comparison

Example	Classification	Risk Level	Main Risk Factors
Example 1	Phishing	High	Brand impersonation, unexpected fax, suspicious link, social engineering
Example 2	Safe / Legitimate-looking	Low	No obvious phishing indicators visible; authenticity not fully verified

Example 3	Suspicious / Potential Phishing	Medium	Unexpected invitation, external call-to-action, possible social engineering
Example 4	Phishing	High	Unexpected valuable offer, upfront fee, personal information request, payment request

7.3 High-Risk Examples

Example 1 and Example 4 were classified as high risk because they contain multiple strong warning signs.

Example 1 combines a trusted-brand appearance with an unexpected document notification and a link designed to encourage interaction.

Example 4 combines an unexpected offer of valuable items with an upfront logistics fee, requests for personal information, and payment options.

7.4 Medium-Risk Example

Example 3 was classified as medium risk because it contains an unexpected invitation and a prominent call-to-action button. However, the available screenshot does not provide enough evidence to conclusively confirm that the email is malicious.

7.5 Low-Risk Example

Example 2 was classified as low risk because the visible PayPal emails did not show obvious phishing indicators. However, the analysis specifically notes that complete verification would require additional information such as email headers, authentication results, and actual URLs.

7.6 Overall Findings

The comparison demonstrates that phishing detection should not depend on a single characteristic. Multiple factors should be considered together, including the sender, context, requested action, links, payment requests, personal-information requests, and social-engineering techniques.

8. PREVENTION AND SECURITY AWARENESS GUIDELINES

Phishing attacks can often be reduced through user awareness and careful handling of unexpected emails. Users should evaluate an email before clicking links, opening attachments, providing information, or making payments.

8.1 Verify the Sender

Users should carefully check the sender's email address and domain. A familiar display name does not necessarily mean that the message was sent by the legitimate organization.

8.2 Avoid Unexpected Links

Users should avoid clicking links or buttons in unexpected emails. When a link is necessary, its destination should be checked before opening it.

For important services, users should preferably access the official website or application directly rather than using an unexpected email link.

8.3 Be Careful With Unexpected Documents

Unexpected fax, document, invoice, or attachment notifications should be treated cautiously. Users should verify the message through a trusted channel before opening or downloading anything.

8.4 Do Not Share Sensitive Information

Users should never provide passwords, authentication codes, financial information, or unnecessary personal information in response to an unexpected email.

Example 2 demonstrates why verification codes should not be shared, while Example 4 demonstrates the risks associated with providing personal information to an unknown sender.

8.5 Be Cautious About Payment Requests

Unexpected requests for money should be treated as a major warning sign.

Users should independently verify any payment request, especially when an email claims that a valuable product, prize, donation, or service is available only after an upfront payment.

8.6 Do Not Be Pressured by Urgency

Attackers may create urgency, exclusivity, or fear to encourage users to act without thinking.

Users should take time to verify unexpected requests instead of responding immediately.

8.7 Verify Through Trusted Channels

If an email appears to come from an organization, users should contact that organization through a known official website, phone number, or application rather than relying on contact information contained in the suspicious email.

8.8 Report Suspicious Emails

Suspicious messages should be reported according to the organization's security or email-reporting procedure.

Reporting helps security teams identify phishing campaigns and protect other users.

8.9 Delete Confirmed Phishing Messages

After reporting a confirmed malicious email, users should delete it and avoid interacting with any remaining links or attachments.

8.10 Quick Phishing Detection Checklist

Before interacting with an unexpected email, users should ask:

- ☐ Do I know the sender?
- ☐ Was I expecting this email?
- ☐ Does the sender's domain look legitimate?
- ☐ Is the email asking me to click a link?
- ☐ Is it asking for personal or financial information?
- ☐ Is it asking me to make a payment?
- ☐ Is it creating unnecessary urgency or pressure?
- ☐ Can I verify the request through an independent trusted channel?

If several answers raise concerns, the email should be treated as suspicious until verified.

9. RECOMMENDATIONS

Based on the analysis of the four email examples, the following recommendations can help improve phishing detection and user security awareness.

9.1 Implement Regular Phishing Awareness Training

Organizations should provide regular awareness training to help users recognize common phishing techniques, including impersonation, suspicious links, unexpected offers, and social engineering.

9.2 Encourage Verification Before Interaction

Users should verify unexpected emails before clicking links, opening attachments, providing information, or making payments.

9.3 Promote Independent Verification

Important requests should be verified using trusted communication channels. Users should not rely solely on contact information or links provided within a suspicious email.

9.4 Strengthen Email Security Controls

Organizations should use appropriate email-security mechanisms such as spam filtering, phishing detection, sender authentication, and malicious-link protection to reduce the number of harmful messages reaching users.

9.5 Establish a Clear Reporting Process

Organizations should provide users with an easy method for reporting suspicious emails. Reported messages can then be investigated by the appropriate security team.

9.6 Encourage a “Think Before You Click” Culture

Users should be encouraged to pause and evaluate unexpected messages rather than immediately responding to requests or clicking links.

9.7 Protect Personal and Financial Information

Users should avoid providing personal, authentication, or financial information through unexpected email requests. Payment requests should always be independently verified.

9.8 Maintain Evidence for Investigation

When a suspicious email is identified, relevant evidence such as screenshots, sender information, message content, and available email headers should be preserved for investigation and reporting.

10. CONCLUSION

The analysis of four different email examples demonstrated that phishing and social-engineering attempts can take many forms. Attackers may imitate trusted services, use unexpected

notifications, create attractive offers, or encourage users to interact with links and provide information.

The examples analyzed in this task showed different levels of risk. Two examples were identified as high-risk phishing emails, one was considered suspicious or potentially phishing, and one appeared legitimate based on the available screenshot evidence, although complete technical verification was not performed.

The analysis demonstrates that users should not judge an email solely by its appearance or branding. Sender information, context, links, requests for information or payment, and social-engineering techniques should all be considered before taking action.

Effective phishing awareness combines careful user behaviour, independent verification, appropriate reporting procedures, and technical email-security controls. Developing these practices can help reduce the likelihood of successful phishing attacks and protect users and organizations from potential security and financial risks.

11. REFERENCES AND RESOURCES

For this project, the primary evidence consisted of the email screenshots and the individual email-analysis documents prepared during the task.

Project Evidence:

1. Example 01 – SharePoint/Fax Notification Email Analysis
2. Example 02 – PayPal Account Notification Analysis
3. Example 03 – Punchbowl Invitation Analysis
4. Example 04 – Free Items / Logistics Fee Analysis
5. Corresponding screenshots stored in the project's Screenshots folder